

Omar Ibne Shahid

Research Interest: AI Security | Pattern Recognition | Mathematical Modeling

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Research Grant

July 2024 - **Mathematical Modeling of Human Activity Recognition in Healthcare Domain**,
Dec 2025 *1 Million BDT*, Funded by Office of Research (OR), North South University.

Professional Research Experience

Mar 2023 - **Translation of a Wearable Fetal Movement Monitor Towards Stillbirth Prevention**,
Dec 2023 **Wellcome Leap In Utero** at *University College Dublin (UCD) (Worked in Hybrid mode))*.
Supervisor: Dr. Niamh Nowlan & Dr. Abhishek Kumar Ghosh,

Oct 2021 - **Machine Learning Based Physical Fatigue Detection System**,
Jan 2023 **The State University of New York at Buffalo**.
Supervisor: Dr. Lora Cavuoto

Jan 2020 - **Human Activity Recognition Using IMU Sensor Unit**,
Dec 2020 **University of East London (Remote)**.
Supervisor: Dr. Md. Atiqur Rahman Ahad

Education

Aug 2021 - **The State University of New York at Buffalo, USA**,
Jan 2023 **MSc in Human Factors Engineering** at **Department of Industrial and System Engineering**
CGPA - 3.50/4.00,
Master's Research: Physical Fatigue Detection; UX Design for Older Adult.

Feb 2016 - **University of Dhaka, Bangladesh**,
Dec 2019 **BSc in Robotics and Mechatronics Engineering**,
CGPA - 3.72/4.00,
Undergraduate Project: Deep Model based Autism Spectrum Disorder (ASD) Detection through Activity Recognition .

Employment History

Mar 2026- **Co-founder & RAG ML Engineer**, at *Leveor*,
Now *A Bangladesh based company, formed LLC in USA..*

Jan 2024- **Lecturer**, *Department of Electrical and Computer Engineering*,
Now *North South University, Dhaka, Bangladesh*,
Courses: Pattern Recognition and Neural Network, Design and Analysis of Algorithm, Digital Logic Design.

Apr 2023- **Research Assistant (Hybrid)**, *WELLCOME LEAP IN UTERO, Dublin, Ireland*,
Dec 2023 **Summary:** Worked on Biomedical Signal Analysis Using Artificial Intelligence to extract fetal movement from the raw signal which is collecting through a wearable device. This is a collaborative project of Imperial College, London, University College Dublin and University of Dhaka..

Aug 2021- **Graduate Teaching Assistant**, *DEPARTMENT OF INDUSTRIAL AND SYSTEM ENGINEERING*,
Dec 2022 *The State University of New York at Buffalo*.

Feb 2020- **Engineer, Level-1**, *DATA SOFT MANUFACTURING & ASSEMBLY INC. LIMITED*,
July 2021 **Embedded System and Artificial Intelligence**.

Jul 2019- **Intern Engineer**, *DATA SOFT MANUFACTURING & ASSEMBLY INC. LIMITED*,
Oct 2019 **Development of Machine Learning and Artificial Intelligence**.

Jul 2018- **Mentor**, *BANGLADESH ROBOT OLYMPIAD NATIONAL TEAM*,
Dec 2019 Coordinated by **Dr. Lafifa Jamal**, Central Committee Member, International Robot Olympiad.
Summary: I was involved in preparing the children for the international round while providing assistance with programming, electrical and mechanical difficulties, as well as debugging their problems. The result of their dedicated efforts was a multitude of awards including gold medal

Research Publication

International Journals

- o "Performance Comparison of Wearable Sensors for Fetal Movement Detection" *IEEE Sensors Journal*, vol. 24, issue no. 17, pp. 28018-28027 [Link](#)

International Conference

- o "Vision-based Action Understanding for Assistive Healthcare: A Short Review (2019, June)." *CVPR Workshops* (pp. 1-11). [Link](#)
- o "UPIC: User and Position Independent Classical Approach for Locomotion and Transportation Modes Recognition (2020, September)." *In Adjunct Proceedings of the 2020 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2020 ACM International Symposium on Wearable Computers* (pp. 340-345). [Link](#)
- o "Classification of Eye-Tracking Data using Saliency Map." *International Conference of Pattern Recognition (ICPR)*, 2021. [Link](#)

Book Chapters

- o "Cooking Activity Recognition with Varying Sampling Rates Using Deep Convolutional GRU Framework." *Human Activity Recognition Challenge*. Springer, Singapore, 2020. (pp. 115-126). [Link](#)
- o "Contactless Human Activity Analysis - Different Modalities." *Contactless Human Activity Analysis (C.H.A.A.)*. Springer, 2020. [Link](#)

Peer Reviewer Experience

- o International Conference on Pattern Recognition (ICPR) 2026
- o ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies 2026
- o Expert Systems with Applications Journal

Honors & Awards

- o Awarded **1st Runner Up** in *Business Idea Innovation Challenge 2017* by BANGLADESH INNOVATION FORUM
- o Honored as **Top 3 Mentors** in *Bangladesh Robot Olympiad 2018* by BANGLADESH OPEN SOURCE NETWORK
- o Awarded **2nd Runner Up** in *Hardware Project Showcase Competition 2019* by ENGINEERING FACULTY, UNIVERSITY OF DHAKA

Projects

- o Pipe Inspection Crawling Robot Using Reinforcement Learning
- o Interbotix Robotic Arm Path Planning Simulation Using Robot Operating System (ROS)
- o Medication Managing UX Interface Design for Older Adult
- o Autistic Child Evaluation System through Their Eye Movement Analysis
- o Finding Optimal Route for Fetching Shelves in a Warehouse Using ROS

All project details can be found at: <https://sites.google.com/view/details-of-previous-projects/home>

Skills

- o Software: C++, Python, Pytorch, TensorFlow, Langchain, Langraph, RAG
- o Hardware: Microcontroller, PLC, 3D Printing, PCB Printing

Organizing and Volunteering Experience

Jan 2025 - **Faculty Advisor**, *IEEE Robotics and Automation Society (RAS)*, NSU Branch.
Present

2018 & 2019 **Organizer**, *Bangladesh Robot Olympiad (National Round)*.

2018 - 2019 **Treasurer**, *IEEE Robotics and Automation Society (RAS)*, Dhaka University Branch.

References

Prof. Niamh C. Nowlan
University College Dublin
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Prof. Brandon Oubre
University of Alabama at Birmingham
boubre@uab.edu

Prof. Md. Atiqur Rahman Ahad
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